

ARYA CHAKRABORTY

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SUMMARY

Data Scientist with experience delivering production ML and GenAI systems in large-scale BFSI environments. Strong background in statistics and applied machine learning, focused on building scalable, data-driven solutions with measurable business impact.

EDUCATION

- Vellore Institute of Technology | M.Sc. Data Science | CGPA: 8.83August 2024
 - Coursework: Python, Data structures, Machine learning, deep learning, Information retrieval techniques.
- Kalyani Mahavidyalaya | B.Sc. Statistics | CGPA: 8.26June 2022
 - Coursework: Hypothesis testing, Probability theory, Time series analysis, Regression analysis.

EXPERIENCE

- Data ScientistDecember 2024 – Present
Tata Consultancy ServicesMumbai - India
 - Delivered an LLM-powered campaign automation platform that converted business PDFs into complex SQL queries, reducing campaign query development time by **approx. 90% (2–3 days to 2–3 hours)** through automated generation and optimization.
 - Designed and deployed an AI-driven website compliance monitoring system combining NLP and Computer Vision to detect broken links, missing disclosures, readability, accessibility gaps, and content inconsistencies, **contributing to 15+ AI use cases** across text classification, image analysis, and structured data extraction.
 - Built a hybrid ResNet-18 based metric learning model for signature verification on around 95k mortgage user signatures, achieving ~80% accuracy and delivering ~\$3M annual savings through automated forgery detection.
 - Built a custom **Computer Vision + OCR document processing pipeline** for mortgage workflows, accelerating document turnaround by 40%+ and improving downstream customer retention by streamlining verification and data extraction stages.
 - Developed and **deployed 13+ predictive models (propensity, response, and segmentation models)** on Azure ML to optimize campaign targeting, improving customer selection precision and campaign ROI.
- AI Developer/ConsultantJanuary 2024 – Nov 2024
Isnartech PVT. LtdGurgaon - India
 - Delivered an end-to-end ERP system integrating **10+ AI features**, including automatic grading, an institutional chatbot, and AI-driven attendance, contributing to a **22% improvement in annual institutional review ratings**.
 - Developed a computer vision-based system for a health & fitness client to estimate calorie intake and body weight from food images.
 - Served as the team lead, guiding and mentoring a team to ensure the successful completion of **more than 4 projects**.
- AI Developer internJanuary 2024 – May 2024
Isnartech PVT. LtdRemote
 - Developed a Flask app for real-time air quality index prediction, integrating advanced ML models (Linear Regressor, Random Forest Regressor, Artificial Neural Network), each achieving over **85% accuracy**, by fetching live air pollutants data.

SKILLS

- Data Analysis & Statistics:** Statistical analysis, Time Series analysis, Probability.
- Programming & Tools:** Python, R, Power Bi, Excel.
- Frameworks:** Transformers, Streamlit, Flask Framework.
- Machine Learning & AI:** Machine Learning, Deep Learning, Artificial Neural Network, Recurrent Neural Network, Convolutional Neural Network, Natural Language Processing (NLP), Computer Vision, Large Language Models, Generative AI (Gen AI).
- Cloud AIML & Version Control:** Azure ML, VPS development, Git.

PROJECTS

1. [Natural Language to SQL Query Conversion using Large Language Model](#)January 2024 – April 2024
 - I fine-tuned Gemma 2B & DeepSeek Coder 1.3B LLMs on 3000 rows of structured data, achieving **SACRE BLEU scores of 70.38 and 70.35** respectively, for converting natural language to SQL queries.
 - The DeepSeek Coder-1.3B base model showed better baseline results compared to GEMMA-2B. After optimization, the model achieved **70.35 SACRE BLEU, 0.703 BLEU**, and an EM ratio of **0.02**, closely matching GEMMA-2B's refined results.
 - The evaluation highlights that both models improved substantially with fine-tuning, with GEMMA-2B slightly outperforming DeepSeek Coder-1.3B in EM ratio.
2. [NLP-Powered Automatic Document Summarization: A Fine-Tuned Approach](#)November 2023
 - Created an Automatic Document Summarization Model using NLP techniques, secured a remarkable **10.71% improvement** in accuracy (**ROUGE score of 0.31**) over pre-trained frameworks.
 - Implemented user-friendly features, allowing customization of summary lengths and support for over 150 languages, ensuring a diverse and efficient user experience.
 - Integrated **4 state-of-the-art LLM models**, including a fine-tuned BART (Facebook/bart-large-cnn), keyword generation using KeyBERT, language translation with Hugging Face Transformers, and LoRA-PEFT for long text summaries.
3. [Gender Classification using CNN & OpenCV](#)July 2023 – September 2023
 - Constructed gender classification system securing **89% accuracy using custom CNN models** on 17,000+ images.
 - Enforced transfer learning and fine-tuning techniques, achieving a **90% test accuracy** on selected model.
 - Designed and trained 5 models for gender classification, including 3 CNNs built from scratch and 2 utilizing transfer learning.

AWARDS

- Tata Consultancy Services
 - Awarded “Special Initiative Award” (TCS AI Hackathon) and “AI Risk Hero Award” for identifying and reducing LLM hallucination issues in a production system.
 - Achieved Runner-Up among 43 global teams at TCS AI Innovista competition.